

TRONDHEIM VOLL, Norway

WMO: 012570

Lat: 63.4106N

Lon: 10.4539E

Elev: 431

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	10.7	-0.4	5.5	9.7	2.8	6.4	13.7	22.6	33.5	20.6	35.0	4.9	210	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	76.6	61.5	73.2	60.1	69.8	58.8	63.6	73.3	61.7	70.6	59.8	67.9	6.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	76.7	67.8	57.5	71.7	65.5	55.9	67.7	63.5	29.1	73.4	27.7	70.8	26.5	67.9	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.5	13.2	DB	2.9	81.5	5.5	3.6	-1.1	84.1	-4.3	86.2	-7.4	88.2	-11.4	90.8	(4)
(5)			WB	2.1	66.8	5.2	2.5	-1.7	68.7	-4.7	70.1	-7.6	71.6	-11.4	73.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	29.1	29.7	33.3	40.9	48.1	53.7	59.3	57.9	51.5	42.9	35.4	31.5
	DBStd	12.71	8.52	8.52	7.21	5.93	7.20	5.95	6.23	5.34	5.29	6.12	7.55	8.46
	HDD50	3454	646	567	519	281	125	25	2	3	43	232	439	572
	HDD65	8114	1111	987	984	723	524	341	200	229	404	685	889	1037
	CDD50	853	0	0	0	8	67	137	291	249	89	12	0	0
	CDD65	38	0	0	0	0	1	3	24	10	0	0	0	0
	CDH74	248	0	0	0	0	7	33	170	38	0	0	0	0
	CDH80	35	0	0	0	0	0	1	33	1	0	0	0	0
Wind	WSAvg	6.0	6.8	6.5	6.9	6.0	5.7	5.6	5.0	4.9	5.5	5.9	6.0	6.8
Precipitation	PrecAvg	32.9	2.6	2.3	2.3	1.9	1.7	2.6	2.6	2.4	4.2	4.0	2.8	3.4
	PrecMax	48.6	5.4	4.6	7.9	3.5	3.6	5.4	4.4	4.6	7.8	6.9	6.4	12.9
	PrecMin	21.3	0.0	0.1	0.1	0.6	0.2	0.5	0.5	0.4	0.7	0.4	0.6	0.8
	PrecStd	8.0	1.4	1.2	1.8	0.8	0.9	1.3	1.2	1.0	1.7	2.0	1.7	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	47.9	51.1	64.8	74.6	78.5	84.0	77.9	69.8	62.0	53.2	48.5
		MCWB	39.6	40.7	41.7	50.1	58.0	60.2	67.2	61.5	56.7	52.4	46.5	42.7
	2%	DB	43.5	44.4	47.0	58.6	69.6	73.0	78.9	74.5	65.6	58.0	50.5	45.9
		MCWB	38.7	38.5	40.5	47.2	55.9	58.2	63.8	61.8	56.1	49.8	44.6	40.2
	5%	DB	41.8	42.1	45.0	54.4	65.3	68.1	74.2	71.0	62.4	54.8	47.5	43.9
		MCWB	37.6	37.1	39.5	45.0	54.2	56.8	61.4	60.1	54.2	48.2	42.3	38.9
	10%	DB	39.8	40.3	42.8	50.2	60.6	64.4	70.6	67.5	59.8	51.7	45.1	42.0
		MCWB	36.3	36.1	37.5	42.7	51.4	55.1	59.9	58.6	52.9	46.2	40.6	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	41.8	44.4	51.7	60.2	62.5	68.5	65.0	59.3	53.8	48.3	44.3
		MCDB	43.5	45.6	47.0	62.4	72.4	73.1	80.6	74.5	66.8	60.8	52.3	46.1
	2%	WB	40.0	39.8	42.1	48.3	57.0	59.7	65.2	62.8	57.1	50.6	45.5	41.8
		MCDB	42.4	42.8	45.8	56.7	67.2	70.1	76.7	71.5	63.0	56.4	49.4	44.7
	5%	WB	38.1	38.2	39.9	45.9	54.7	57.7	62.7	61.1	55.4	48.9	43.2	39.6
		MCDB	40.9	41.3	44.1	52.7	64.1	66.1	72.1	69.2	60.8	54.0	46.6	42.8
	10%	WB	36.4	36.3	38.1	43.6	51.8	55.7	60.7	59.2	53.7	47.0	41.0	37.9
		MCDB	39.5	39.7	42.1	49.4	60.7	63.2	68.9	65.9	58.7	51.3	44.7	41.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.6	10.7	12.2	13.0	12.6	13.1	12.6	10.9	9.6	8.1	8.4
		MCDBR	8.3	8.8	12.0	18.3	20.8	19.8	20.1	18.1	15.0	12.5	10.8	10.0
		MCWBR	5.9	6.4	8.0	10.3	10.6	9.7	9.2	8.8	8.3	7.8	7.5	7.6
	5% WB	MCDBR	7.8	8.3	10.9	16.3	19.6	17.7	19.0	16.6	14.2	10.9	10.5	9.6
		MCWBR	6.1	6.7	8.0	9.7	10.6	9.4	9.3	8.7	8.5	7.4	7.9	7.8
Clear-Sky Solar Irradiance	taub	0.223	0.285	0.314	0.341	0.344	0.345	0.354	0.342	0.322	0.300	0.365	0.355	
	taud	2.043	2.237	2.290	2.372	2.447	2.478	2.474	2.503	2.545	2.457	2.732	2.655	
	Ebn at Noon	125	206	247	267	276	278	272	265	249	205	119	69	
	Edh at Noon	9	19	26	30	31	30	30	27	21	15	8	5	
All-Sky Solar Radiation	RadAvg	50	219	620	1185	1453	1513	1466	1125	677	299	79	24	
	RadStd	8	33	88	94	97	153	150	130	70	40	13	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page